

Digital Evidence

Week 9 – Case Study: From Beginning to Trial

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ITMS X83 – Spring 2026

Slides contain original content from Davis, S. and
E-discovery: An Introduction to Digital Evidence



Last Week - Review

- Kahoot Review – 16 questions
 - 30 seconds for multiple choice
 - 20 seconds for true and false

Objectives

- Describe prelitigation e-discovery investigation practices
- Describe common e-discovery documents used when starting a formal lawsuit
- Explain e-discovery project management tasks, including sampling and metrics
- Explain fact-finding and pretrial motions

Overview

1. Early Case Evaluation
2. Starting Litigation
3. Plaintiff's E-discovery Management
4. E-discovery Organization
5. Pretrial Motions & Trial

Part I

Early Case Evaluation

Early Case Evaluation

- Establishing a basis of a lawsuit involves early fact finding
 - To lay a foundation for the initial claims
- Not unusual for a case's basis to expand as investigation and discovery yield more information

Prelitigation Investigations

- Before a complaint or indictment is filed:
 - A preliminary investigation should reveal some of the factual foundation for the litigation
- In civil cases:
 - Client is interviewed and any preliminary evidence is evaluated
- In criminal cases:
 - Covert actions such as wiretaps and body wires might occur

Prelitigation Investigations

- During the investigation stage:
 - Goal is to capture as much potential evidence as possible
 - Before litigation and before a legal hold or notifying the opposition of potential litigation

Prelitigation Investigations

- Let's say Widget Corp had a recent breach and you were investigating their security posture
- What type of tools and info would you use/look for that are available publicly online??

Prelitigation Investigations

- Prelitigation investigations in general might include:
 - Capturing Web sites, YouTube video, social networking profiles, information from government or media resources, geospatial or mapping data, webcam video, cell phone or voicemail recordings, etc.

Prelitigation Investigations

- More advanced investigative tools include analytic features
 - They are marketed for law enforcement use but are also used in civil litigation
 - Examples: Analyst's Notebook and Social Network Analyzer
- These tools create visual representation of large data sets such as:
 - Phone or bank records, IP address and traffic, social networking diagrams, link analysis, geospatial analysis, and tree maps

Reconnaissance Investigations Tools

- Some tools I commonly use for reconnaissance and investigation are:
 - Centralops.net
 - Shodan
 - Censys
 - DomainTools
 - DNS Dumpster
 - Maltego
 - Google

Fictional Investigation

- Let's say we were investigating the Percussive Arts Society to see how many servers and subdomains it has online

*Disclaimer. We are solely performing passive reconnaissance. You do **not** have permission to port scan or maliciously interact with any of the sites we look at or uncover

Central Ops.net

Central Ops.net *Advanced online Internet utilities*

Utilities

- Domain Dossier
- Domain Check
- Email Dossier
- Browser Mirror
- Ping
- Traceroute
- NsLookup
- AutoWhois
- AnalyzePath

Free online network tools

Tools

Domain Dossier

Investigate domains and IP addresses. Get registrant information.

enter a domain or IP address

or [learn about yourself](#)

Domain Check

See if a domain is available for registration.

Email Dossier

Validate and troubleshoot email addresses.

Browser Mirror

See what your browser reveals about you.

Ping

See if a host is reachable.

CentralOps.net

- Enter pas.org and hit “go”
- What is the IP address of the webserver?
- What network range is the server in?

Address lookup

canonical name **pas.org.**

aliases

addresses **173.248.172.34**

Queried whois.arin.net with "n 173.248.172.34"...

NetRange: 173.248.128.0 - 173.248.191.255

CIDR: 173.248.128.0/18

NetName: NET-WEHOST-3

NetHandle: NET-173-248-128-0-1

Parent: NET173 (NET-173-0-0-0-0)

NetType: Direct Allocation

OriginAS: AS30475

Organization: Handy Networks, LLC (WEHOST-1)

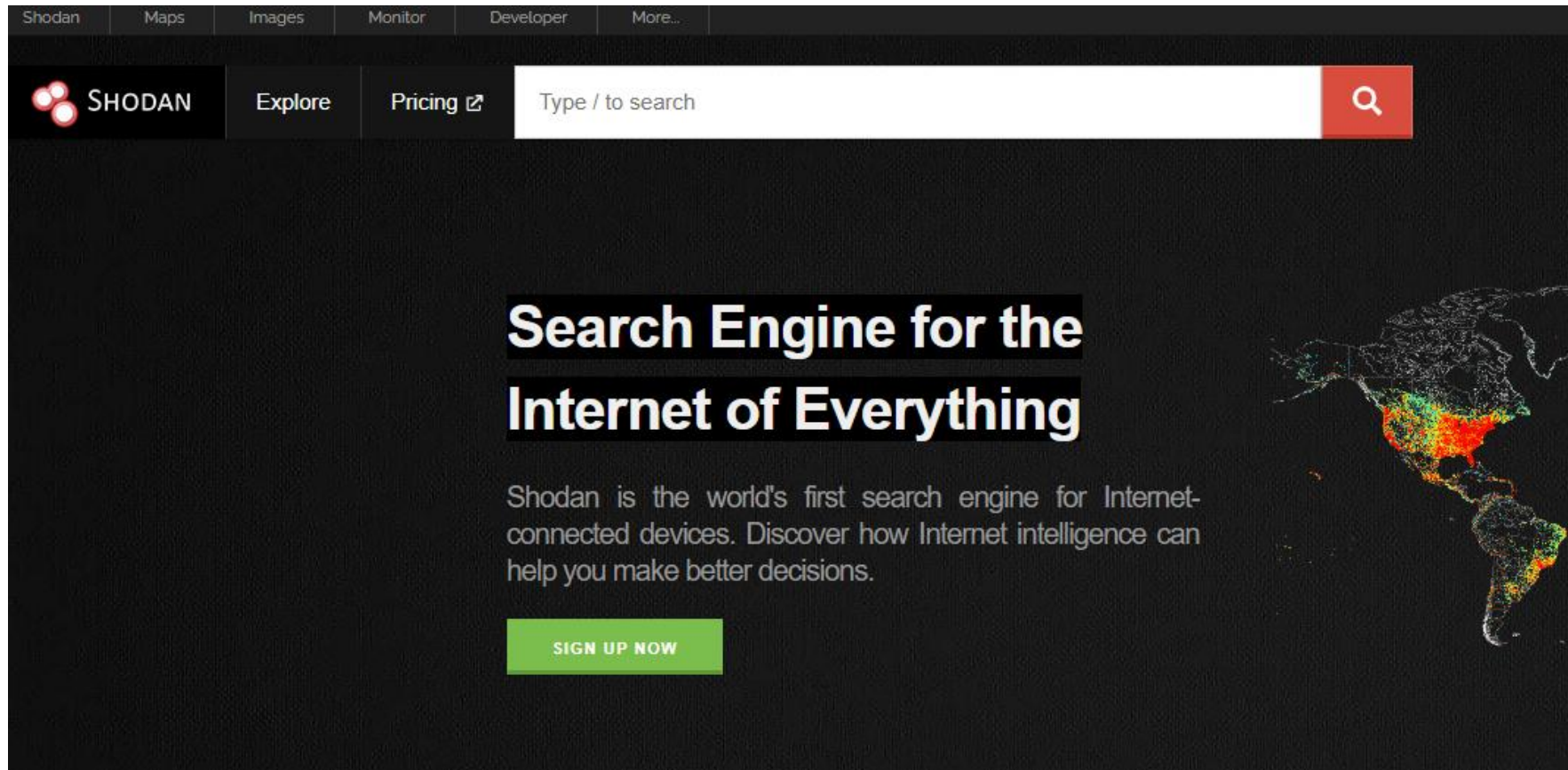
RegDate: 2010-05-13

Updated: 2013-02-11

Ref: <https://rdap.arin.net/registry/ip/173.248.128.0>

Shodan

- Go to shodan.io

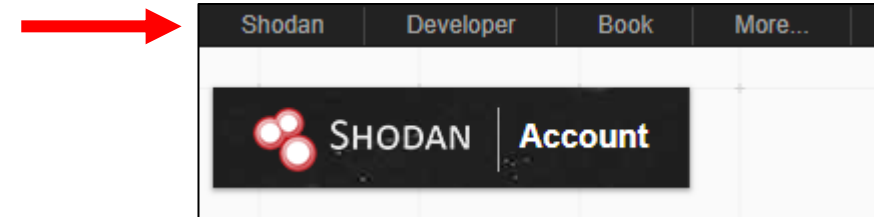


Shodan

- Register for an account by clicking on “Login” on the upper right
- Now click to red “Register” button
- Pick a username and password, provide your email, and hit “Create”
- Check your email to activate the account

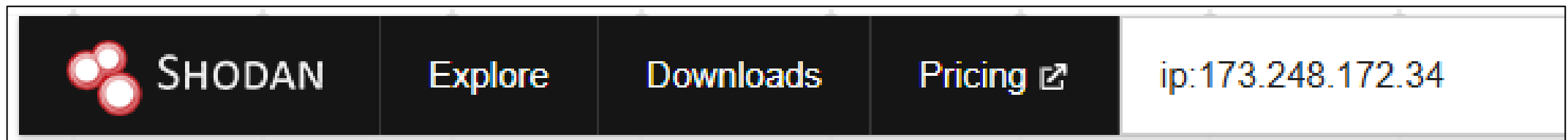
Shodan

- After clicking on the activation link, enter your username and password
- Select Shodan to go back to the home screen



Shodan

- Enter the PAS IP address in the search bar after using ip: and hit Enter:



- Click on the top entry:

173.248.172.34
173-248-172-34.cprapid.com
cpcontacts.173-248-172-34.cprapid.com
whm.173-248-172-34.cprapid.com
cpanel.173-248-172-34.cprapid.com
autodiscover.173-248-172-34.cprapid.com
Softsys Hosting USA, Inc
United States, Denver

SSL Certificate
Issued By:
|- Common Name:
R10
|- Organization:
Let's Encrypt
Issued To:
|- Common Name:
173-248-172-34.cprapid.com
Supported SSL Versions:
TLSv1.2, TLSv1.3

Shodan

- How many ports are open on the server?
 - 13
- What services are running on these ports?
 - 21 FTP
 - 53 PowerDNS
 - 80/443: Apache
 - 110/993/995: Dovecot (POP3/IMAPS/POP3S)
 - 465/587: Exim SMTP
 - 2082/2083/2086/2087: cPanel GUI mgmt

Shodan

- Who is the hosting provider?
 - Softsys Hosting USA, Inc.
- What web server software do they use?
 - Apache

Shodan

- Someone told us an FTP server may have been running with an open port at one point.
- When was that last running and how did you find out?
 - Select port 21 and you'll see on the far right in small gray print that Pure-FTPd last ran on 2/23/26

Question

- If you only know the IP of the server, can you easily see all of the websites/pages on that server??
 - No
- Why??
 - Most servers have virtual hosts where many websites are run off of a single server

Question

- How do you find out what other sites are running on a server??
 - Actively: An attacker or pen tester could scan the public facing server with a wordlist through a dictionary or brute force scan
 - Don't do that on any site without written permission!
 - How would you figure this out passively??
 - Use sites like Censys or DomainTools!

Censys

Censys

Q IPv4 Hosts ip:174.143.128.0/22

[Results](#) [Map](#) [Metadata](#) [Report](#)

Quick Filters

For all fields, see [Data Definitions](#)

Autonomous System:

- 181 RMH-14 - Rackspace Hosting

Protocol:

- 124 22/ssh
- 95 80/http
- 79 443/https
- 17 25/smtp
- 14 21/ftp

[More](#)

Tag:

- 124 ssh
- 114 http
- 77 https
- 20 database
- 17 smtp

[More](#)

IPv4 Hosts

Page: 1/8 Results: 181 Time: 204ms Query Plan: [expanded](#)

174.143.130.194 (management.dialsource.com) 🌐 RMH-14 - Rackspace Hosting (33070) 📍 San Antonio, Texas, United States 🐧 Ubuntu ⚙️ 22/ssh
174.143.130.48 🌐 RMH-14 - Rackspace Hosting (33070) 📍 San Antonio, Texas, United States 🐧 Ubuntu ⚙️ 22/ssh
174.143.130.167 (www-c.iit.edu) 🌐 RMH-14 - Rackspace Hosting (33070) 📍 San Antonio, Texas, United States 🐧 Ubuntu ⚙️ 443/https, 80/http 🏠 Blogs at IIT 🔒 *.iit.edu, iit.edu
174.143.130.18 🌐 RMH-14 - Rackspace Hosting (33070) 📍 San Antonio, Texas, United States 🐧 Ubuntu ⚙️ 443/https, 80/http 🏠 Mayfield 🔒 www.bacchusmanagement.com, bacchusmanagement.com

Censys

The screenshot shows the Censys search interface. At the top, the Censys logo is on the left, and a search bar on the right contains the text 'Certificates' and 'iit.edu'. Below the search bar, the page is divided into two main sections: 'Quick Filters' on the left and 'Certificates' on the right. The 'Quick Filters' section includes a 'Tag' filter with options like 'CT', 'Google CT', 'Leaf', 'Expired', and 'Previously Trusted', and an 'Issuer' filter with options like 'Let's Encrypt', 'DigiCert Inc', 'GoDaddy.com, Inc.', 'Illinois Institute of Technology', and 'COMODO CA Limited'. The 'Certificates' section displays a list of search results. The first result is highlighted with a red box and shows the following details: 'C=US, ST=Illinois, L=Chicago, O=Illinois Institute of Technology, CN=*.iit.edu', 'DigiCert SHA2 High Assurance Server CA', '2016-08-18 - 2019-11-16', and 'parsed.extensions.subject_alt_name.dns_names: *.iit.edu, cmcadmin.iit.edu, hipi.iit.edu, iit.edu, ...'. The second result is also highlighted with a red box and shows: 'C=US, ST=Illinois, L=Chicago, O=Illinois Institute of Technology, CN=*.iit.edu', 'DigiCert SHA2 High Assurance Server CA', '2019-07-30 - 2019-11-16', and 'parsed.extensions.subject_alt_name.dns_names: *.iit.edu, iit.edu, myparking.iit.edu'. The third result shows: 'C=US, ST=Illinois, L=Chicago, O=Illinois Institute of Technology, OU=Office of Technology Services, CN=*.iit.edu', 'DigiCert SHA2 High Assurance Server CA', '2019-09-11 - 2021-11-21', and 'parsed.extensions.subject_alt_name.dns_names: *.iit.edu, iit.edu'.

Censys

[Results](#) [Report](#)

Quick Filters
For all fields, see [Data Definitions](#)

Tag:

- 1,238 CT
- 1,237 Google CT
- 1,201 Leaf
- 1,038 Expired
- 844 Previously Trusted
- [More](#)

Issuer:

- 673 Let's Encrypt
- 364 DigiCert Inc
- 80 GoDaddy.com, Inc.
- 24 Illinois Institute of Technology
- 20 COMODO CA Limited
- [More](#)

Certificates
Page: 1/62 Results: 1,547 Time: 645ms

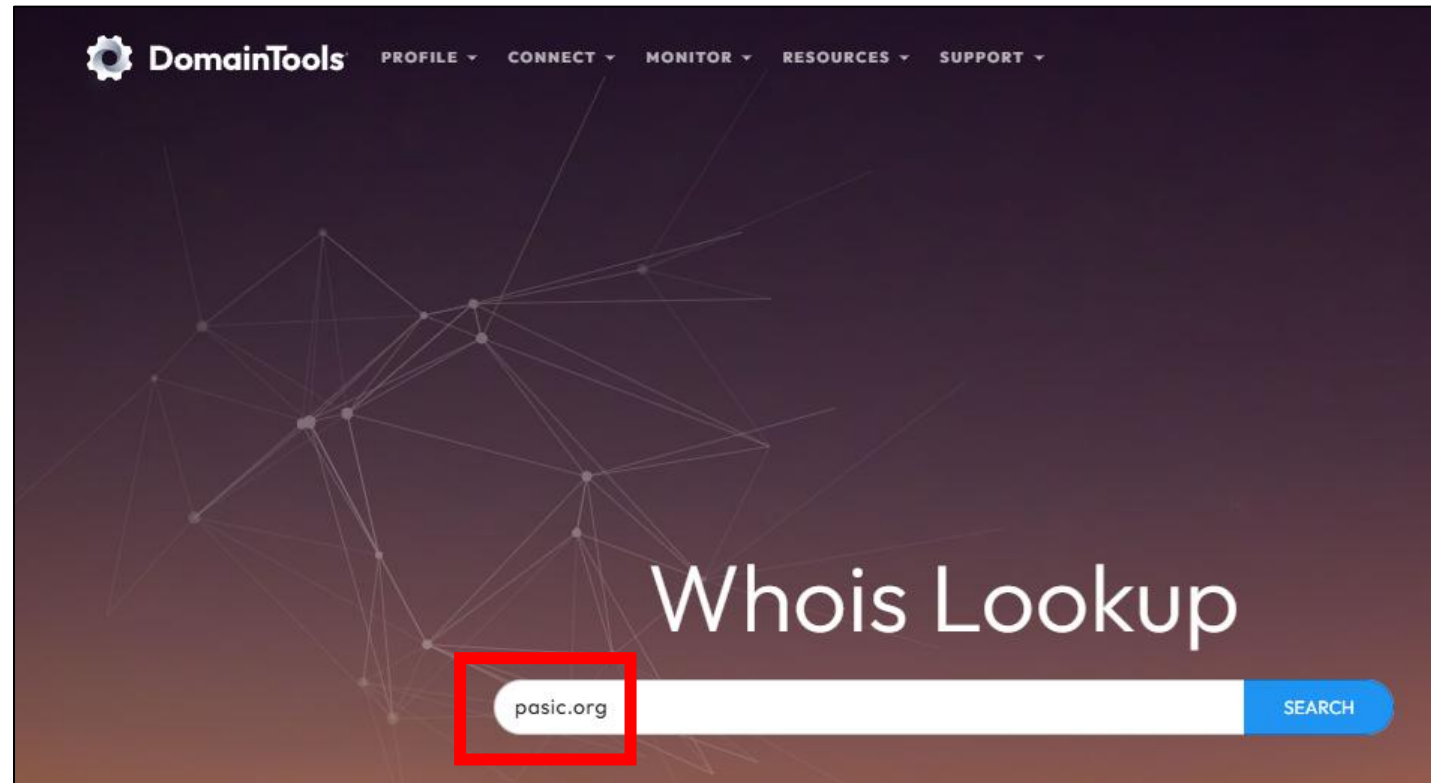
C=US, ST=Illinois, L=Chicago, O=Illinois Institute of Technology, CN=*.iit.edu
 DigiCert SHA2 High Assurance Server CA
 2016-08-18 – 2019-11-16
 *.iit.edu, cmcadmin.iit.edu, hipi.iit.edu, iit.edu, ...
 parsed.extensions.subject_alt_name.dns_names: **iit.edu**

C=US, ST=Illinois, L=Chicago, O=Illinois Institute of Technology, CN=*.iit.edu
 DigiCert SHA2 High Assurance Server CA
 2019-07-30 – 2019-11-16
 *.iit.edu, iit.edu, myparking.iit.edu
 parsed.extensions.subject_alt_name.dns_names: myparking. **iit.edu**

C=US, ST=Illinois, L=Chicago, O=Illinois Institute of Technology, OU=Office of Technology Services, CN=*.iit.edu
 DigiCert SHA2 High Assurance Server CA
 2019-09-11 – 2021-11-21
 *.iit.edu, iit.edu
 parsed.extensions.subject_alt_name.dns_names: **iit.edu**

DomainTools

- Go to whois.domaintools.com
- Search for pasic.org



DomainTools

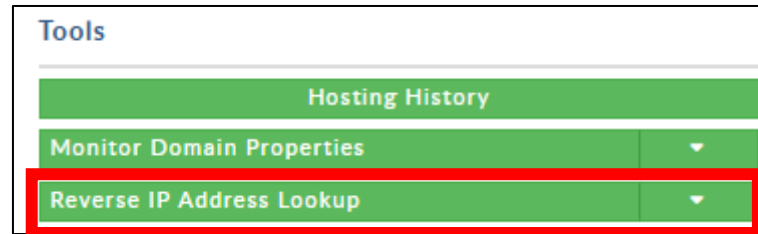
Whois Record for Pasic.org

— Domain Profile

Proximity Score	15	More Info ➞
Email	yx43t6h...@networksolutionsprivateregistration.com domain.operations@web.com is associated with ~10,207,671 domains	➞
Registrar	Register.com - Network Solutions, LLC IANA ID: 9 URL: http://www.register.com Whois Server: whois.register.com domain.operations@web.com (p) +1.8777228662	
Registrar Status	clientTransferProhibited	
Dates	9,847 days old Created on 1999-03-22 Expires on 2027-03-22 Updated on 2026-02-20	➞
Name Servers	NS1.PASIC.ORG (has 1 domains) NS2.PASIC.ORG (has 1 domains)	➞
IP Address	173.248.172.34 - 2 other sites hosted on this server	➞
IP Location	🇺🇸 - Colorado - Denver - Handy Networks Llc	
ASN	🇺🇸 AS30475 WEHOSTWEBSITES-COM - Handy Networks, LLC, US (registered Oct 16, 2003)	
Domain Status	Registered And Active Website	
Whois History	100 records have been archived since 2001-04-03	➞
IP History	16 changes on 16 unique IP addresses over 21 years	➞
Hosting History	5 changes on 5 unique name servers over 11 years	➞

DomainTools

- Perform a reverse IP search by clicking on the arrow to the right of “IP Address”



- What domains are on this server?
 - pas.org
 - pasic.org
 - rhythmdiscoverycenter.org

DomainTools

- The pas.org site is an anonymously registered domain.

Whois Record (last updated on 2026-03-07)

```
Domain Name: PASIC.ORG
Registry Domain ID:
Registrar WHOIS Server: whois.register.com
Registrar URL: http://www.register.com
Updated Date: 2026-02-20T07:58:49Z
Creation Date: 1999-03-22T05:00:00Z
Registrar Registration Expiration Date: 2027-03-22T05:00:00Z
Registrar: Register.com - Network Solutions, LLC
Registrar IANA ID: 9
Reseller:
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Registry Registrant ID:
Registrant Name: PERFECT PRIVACY, LLC
Registrant Organization:
Registrant Street: 5335 Gate Parkway care of Network Solutions PO Box 459
Registrant City: Jacksonville
Registrant State/Province: FL
Registrant Postal Code: 32256
Registrant Country: US
Registrant Phone: +1.5707088622
```

DomainTools

- How can you potentially find out who the real or original registrant was??
 - Look at historical snapshots of Whois info!
 - DomainTools has this ability but only for paid accounts.

DomainTools


DomainTools

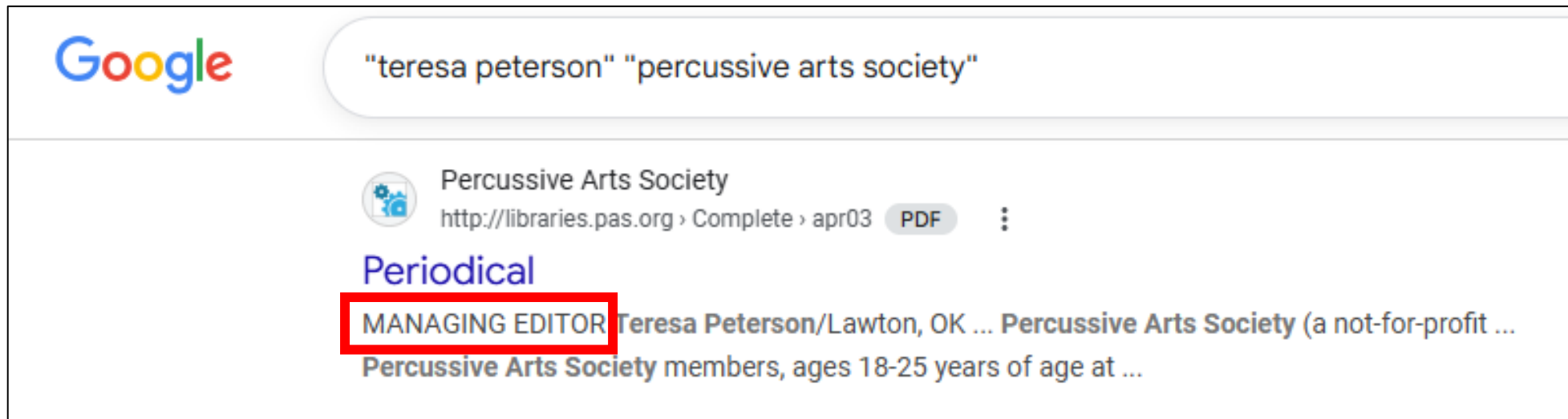
[PROFILE ▾](#)
[CONNECT ▾](#)
[MONITOR ▾](#)
[RESOURCES ▾](#)
[SUPPORT ▾](#)

2012 5 total show additional records up to 2014-01-11 2011 6 total show additional records up to 2014-01-11 2011-10-12 more changes screenshot 2011-05-28 more changes screenshot 2010 8 total show additional records up to 2011-05-28 2010-02-03 more changes screenshot 2009 8 total show additional records up to 2010-02-03 2008 5 total show additional records up to 2010-02-03 2007 2 total show additional records up to 2010-02-03 2003 1 total 2003-11-24 more changes screenshot 2001 1 total > 2001-04-03 more changes screenshot	teresa@pas.org 🔍 jjohn@sirinet.net 🔍 Registrant: Percussive Arts Society 701 NW Ferris Ave Lawton, OK 73507 US Domain Name: PASIC.ORG Administrative Contact: Peterson, Teresa teresa@PAS.ORG Percussive Arts Society 701 NW Ferris Ave Lawton , OK 73507 355-4468 Technical Contact: Johnson, John jjohn@SIRINET.NET Sirius Systems Group, Inc. 3102 Cache Road Suite 101 Lawton, OK 73505 US 580-355-6436 Fax- 580-351-0115 Record last updated on 03-Apr-2001 Record expires on 22-Mar-2003 Record created on 22-Mar-1999 Database last updated on 11-Jan-2002 15:00:24 EST Domain servers in listed order: <table border="0" style="width: 100%;"> <tr> <td>NS1.SIRINET.NET</td> <td>198.203.196.65</td> </tr> <tr> <td>NS2.SIRINET.NET</td> <td>198.203.196.66</td> </tr> </table>	NS1.SIRINET.NET	198.203.196.65	NS2.SIRINET.NET	198.203.196.66
NS1.SIRINET.NET	198.203.196.65				
NS2.SIRINET.NET	198.203.196.66				

[scroll for more records](#)

DomainTools

- Using the internet, how was Teresa Peterson related to the Percussive Arts Society?

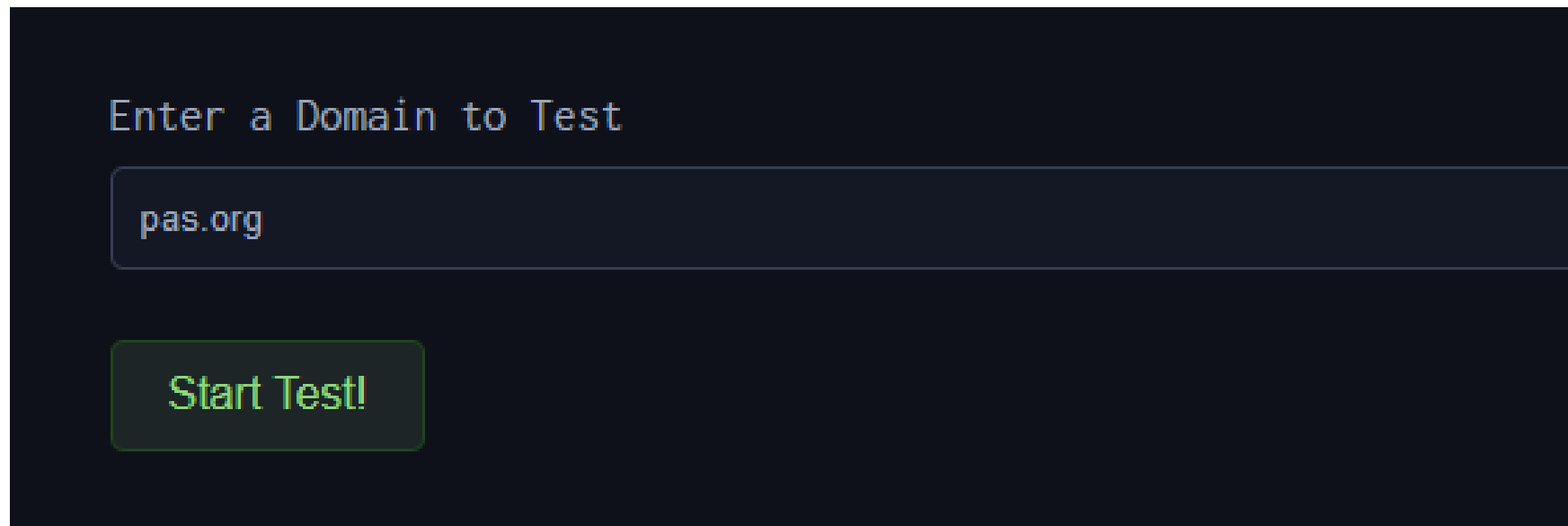


DNS Dumpster

- Used to find subdomains
- Why would subdomains be useful??
 - Test pages not yet live
 - Pages that are “secured” only by obscurity
 - Development APIs
 - Front ends to web applications (may be unsecured)
 - Etc.

DNS Dumpster

- Go to dnsdumpster.com
- Enter pas.org and hit “Start Test!”



Enter a Domain to Test

pas.org

Start Test!

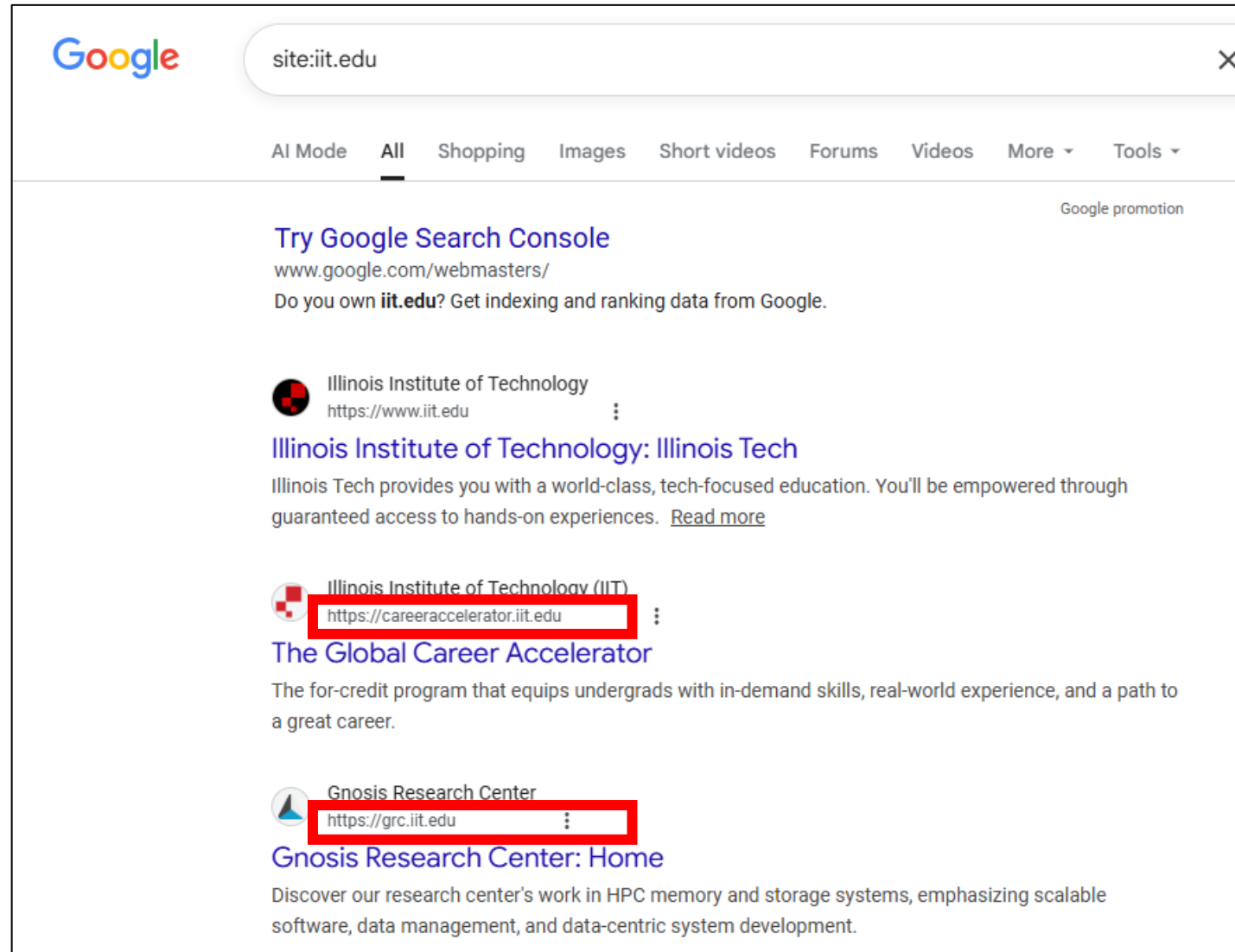
DNS Dumpster

- What information are you provided for the “A Records” (AKA the subdomains)?
 - Host/Subdomain
 - IP address
 - Network assignment info
 - Open services (from a DB, not a port scan)
- How many “A” records were found?
 - 19

Questions

- Will all subdomains uncovered always be publicly accessible?
 - No, some may be internally facing only
- What is another way to find subdomains as well as find all of the publicly available documents for a domain or subdomain??
 - Google's site search operator!

Google



Google

site:iit.edu

AI Mode All Shopping Images Short videos Forums Videos More Tools

Google promotion

[Try Google Search Console](#)
www.google.com/webmasters/
Do you own **iit.edu**? Get indexing and ranking data from Google.

 Illinois Institute of Technology
https://www.iit.edu

Illinois Institute of Technology: Illinois Tech
Illinois Tech provides you with a world-class, tech-focused education. You'll be empowered through guaranteed access to hands-on experiences. [Read more](#)

 Illinois Institute of Technology (IIT)
https://careeraccelerator.iit.edu

The Global Career Accelerator
The for-credit program that equips undergrads with in-demand skills, real-world experience, and a path to a great career.

 Gnosis Research Center
https://grc.iit.edu

Gnosis Research Center: Home
Discover our research center's work in HPC memory and storage systems, emphasizing scalable software, data management, and data-centric system development.

Google Dorks

- This blog gives a good overview of Google's search operators:
 - <https://ahrefs.com/blog/google-advanced-search-operators/>
- Go ahead and pull it up

Google

- What is a single google search to find the following for only cs.iit.edu:
 - Only pdfs
 - The year “2015” is in the web page url
 - The word “haswell” is in the web page title
- `site:cs.iit.edu filetype:pdf inurl:2015 intitle:haswell`

Google

- What is the name of the single document returned??
 - HPC Power Management on Haswell CPU Architecture
- Let's look at this in stages...

Google

- How many results for each?
- Hit the “Tools” dropdown in Google to see the number of results
 - site:cs.iit.edu
 - 5,650
 - site:cs.iit.edu filetype:pdf
 - 2,910
 - site:cs.iit.edu filetype:pdf inurl:2015
 - 10
 - site:cs.iit.edu filetype:pdf inurl:2015 intitle:haswell
 - 1

Google



site:cs.iit.edu filetype:pdf inurl:2015 intitle:haswell



AI Mode

All

Shopping

Images

Videos

Forums

Short videos

More ▾

Tools ▾



Illinois Institute of Technology (IIT)

[https://datasys.cs.iit.edu › grants › 2015 › reports](https://datasys.cs.iit.edu/grants/2015/reports)

PDF

HPC Power Management on Haswell CPU Architecture

As science continues to generate large sets of data, society expects nothing less than a machine capable of performing intensive calculations in a matter of.

Google Hacking Database

- Another useful resource containing various Google Dorks:
 - <https://www.exploit-db.com/google-hacking-database>

X Advanced Search

× **Advanced search** Search

Words

All of these words
Example: what's happening · contains both "what's" and "happening"

This exact phrase
Example: happy hour · contains the exact phrase "happy hour"

Any of these words
Example: cats dogs · contains either "cats" or "dogs" (or both)

None of these words
Example: cats dogs · does not contain "cats" and does not contain "dogs"

These hashtags
Example: #ThrowbackThursday · contains the hashtag #ThrowbackThursday

Hootsuite

privacy AND ("clas...

Search JoeStuc...

mellizzenn retweeted

Gabriel Mwangi @gabukaratu

3 hours ago

We need to launch class Action Lawsuits against @SafaricomPLC

1. Poor internet

2. Data privacy/ Abductions

2

1

Gabriel Mwangi @gabukaratu

3 hours ago

We need to launch class Action Lawsuits against @SafaricomPLC

1. Poor internet

2. Data privacy/ Abductions

2

1

Elonprivate6213 retweeted

American714 @AndrewBeam19

4 hours ago

@elonmusk Have you ever taken a US Civics Class? Have you ever read the Constitution?

Have you ever been sued in a class action suit in State Court for Privacy Laws violations? It's coming soon to a court near you

1

3

In reply to elonmusk

smart device AND ...

Search JoeStuc...

Obinna E N @ObisDev

3 days ago

iPhones! And the best part? Everything stays on your device, so your privacy is safe in your hands. This is a total game-changer for both Apple and Google—smart strategy and a win for users! What do you think about this wild partnership?

#Apple #Google #AI #TechNews #Innovation

2

1

Show Conversation

puppets @Piuroy738

3 days ago

IoT & Edge Computing:

Zama enables secure AI processing on IoT devices, like sensors or smart home devices. Data can be processed directly on the device, without needing to send raw data to the cloud, ensuring privacy and low-latency responses.

Maltego – Infrastructure

The screenshot displays the Maltego Infrastructure interface. On the left is the **Entity Palette** with categories: Recently Used, Infrastructure, and Locations. The **Infrastructure** category is expanded, showing entities like AS, Banner, DNS Name, Domain, IPv4 Address, MX Record, NS Record, Netblock, URL, Tracking Code, Website, Shodan ICS IPv4Address, Airport, Church, Circular Area, and City. The main workspace shows a **New Graph (1)** with a central **array** entity (NS icon) connected to various domain entities like **prop-fitsolutions.ca**, **patersonhousingauthority.com**, **womenwantjewelry.com**, **moodoo.cc**, **fresca.com.au**, **walkthejourney.ca**, **eluxury.bg**, **tehhold.ru**, and **j2ee.com.cn**. A **maltego.000webhostapp.com** entity is also present. A **Run Transform...** dialog is open, showing a list of transforms under the **All Transforms** tab. The transform **To DNS Name - MX (mail server)** is selected, with a tooltip stating: "This transform will find the MX record". The **Output - Transform Output** pane at the bottom shows the results of the transform, including rate limit information and a list of domains returned by the DNSDB lookup.

Entity Palette

- Search:
- Recently Used
- Domain: An internet domain
- Infrastructure
 - AS: An Internet Autonomous System (AS)
 - Banner: Banner
 - DNS Name: Domain Name System server name
 - Domain: An internet domain
 - IPv4 Address: An IP version 4 address
 - MX Record: A DNS mail exchange record
 - NS Record: A DNS name server record
 - Netblock: A range of IP version 4 addresses
 - URL: An Internet Uniform Resource Locator (URL)
 - Tracking Code: Represents a tracking code for a web serv
 - Website: An internet website
 - Shodan ICS IPv4Address: Shodan ICS IP Address
- Locations
 - Airport: A complex of runways and buildings for th
 - Church: A place of worship
 - Circular Area: A circular area somewhere on Earth
 - City

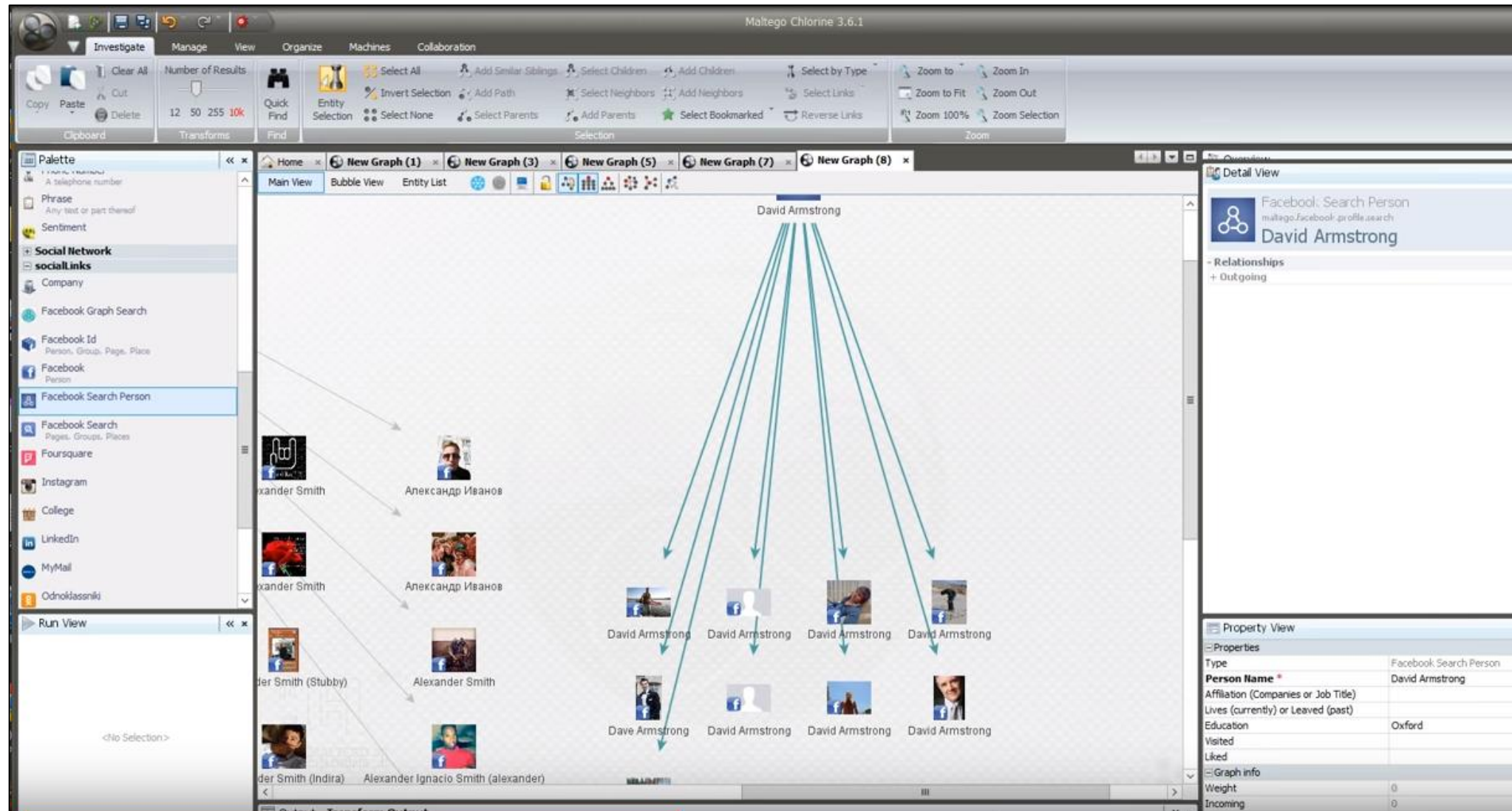
Run Transform...

- [DNSDB] Lookup \$domain.*/A
- [DNSDB] Lookup \$domain.*/AAAA
- [DNSDB] Lookup \$domain.*/CNAME
- [DNSDB] Lookup *. \$domain/CNAME
- [DNSDB] Lookup MX for this Domain
- [DNSDB] Lookup NS for this Domain
- [DNSDB] To DNSNames with this value
- [DNSDB] To records with this hostname
- Enrich breached domain [v2 @haveibeenp...
- To DNS Name (interesting) [Robtex]
- To DNS Name - MX (mail server)
- To DNS Name - NS (name server)
- To DNS Name - SOA (Start of Authority)

Output - Transform Output

```
X-RateLimit-Limit: unlimited
X-RateLimit-Remaining: n/a
X-RateLimit-Reset: n/a (from entity "array")
Free Farsight transforms: 5 of 12 runs over the last 1 h. Buy more: http://integration.paterva.com/Farsight/buy/742cae751a15dbld7118 (from entity "array")
Transform [DNSDB] Domains using this NS returned with 12 entities (from entity "array")
Transform [DNSDB] Domains using this NS done (from entity "array")
X-RateLimit-Limit: unlimited
X-RateLimit-Remaining: n/a
X-RateLimit-Reset: n/a (from entity "array")
```


Maltego – Social Media



Other Useful Tools

- Wayback Machine
 - Find older versions of sites
- Go to web.archive.org
 - Search for www.thefacebook.com
 - Look at the 2/12/2004 snapshot

Other Useful Tools



Other Useful Tools

- EXIFtool
 - Extracting metadata from photos and documents that are publicly available
- LinkedIn and Facebook
 - To view information about employees at a company
- Threat Intelligence Sources
 - Information about malware tied to IPs, URLs, Email..

ThreatConnect

The screenshot displays the ThreatConnect web interface. On the left, a sidebar contains navigation options: Indicators (with a list of types like Address, E-mail Address, File, Host, URL, ASN, CIDR, Mutex, Registry Key, User Agent, each with a checkmark), Groups (with a list of types like Adversary, Campaign, Document, E-mail, Event, Incident, Intrusion Set, Report, Signature, Task, Threat), and Tags. The main content area is titled 'MY THREATCONNECT (54)' and shows a search for 'prennera.com'. A 'FILTERS' dropdown is set to 'prennera.com'. Below the search bar, a table lists indicators for 'prennera.com' with columns for Type and Host. The 'Host' column shows 'prennera.com'. An 'EXPORT' button is visible. The right pane shows details for 'prennera.com', including a status set by icon, a table of source information, and a description. The description states: 'Likely malicious APT C2 domain that resolved to the IP address 142.91.76[.]134 on December 11, 2013, the same date a malicious HttpBrowser APT sample MD5: 02FAB24461956458D70AEED1A028EB9C was observed that connected to that IP. This domain is likely impersonating healthcare provider Premera Blue Cross.' Below this, there are sections for 'Associated Intel' and 'Associated Indicators'.

ThreatConnect

MY THREATCONNECT (54)

FILTERS prennera.com

Contains: prennera.com

Indicators

- Address ✓
- E-mail Address ✓
- File ✓
- Host ✓
- URL ✓
- ASN ✓
- CIDR ✓
- Mutex ✓
- Registry Key ✓
- User Agent ✓

Groups

- Adversary
- Campaign
- Document
- E-mail
- Event
- Incident
- Intrusion Set
- Report
- Signature
- Task
- Threat

Tags

Type

Host prennera.com

EXPORT

prennera.com

Status set by

Source	06-01-2018	Test
Source	02-27-2015	Imported from ThreatConnect Passive DNS
Description	02-27-2015	Likely malicious APT C2 domain that resolved to the IP address 142.91.76[.]134 on December 11, 2013, the same date a malicious HttpBrowser APT sample MD5: 02FAB24461956458D70AEED1A028EB9C was observed that connected to that IP. This domain is likely impersonating healthcare provider Premera Blue Cross.

Associated Intel

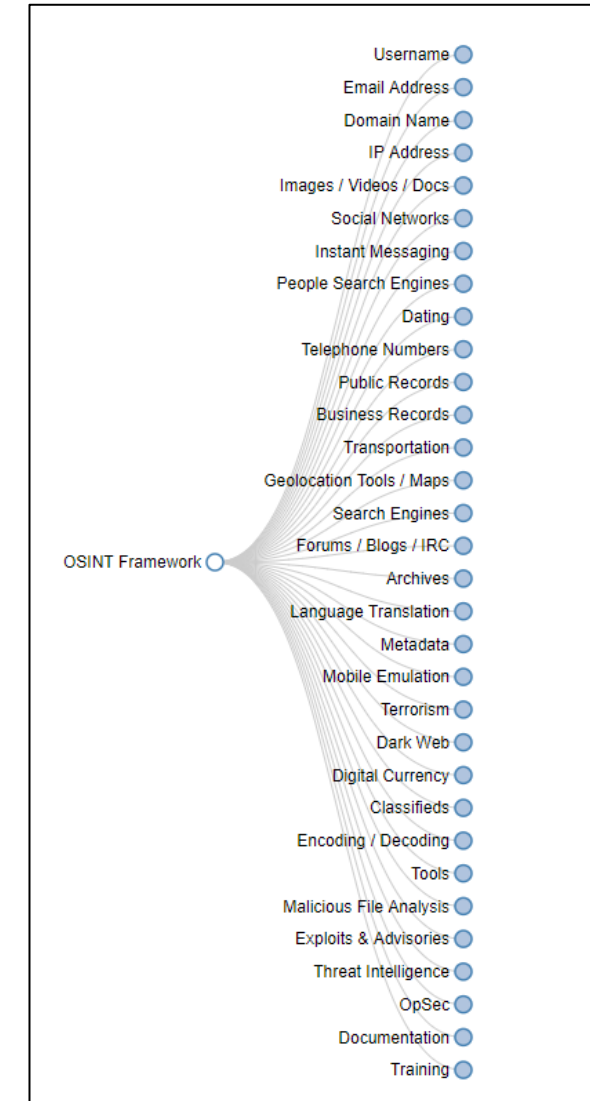
Type	Owner	Date Added
Incident 20131211C: prennera DTOPTOOLZ APT	Common Community	02-27-2015
Incident 20150609B: Blue Cross Blue Shield Targeting I...	Common Community	06-09-2015

Associated Indicators

Type	Owner	Date Added
File 02FAB24461956458D70AEED1A028EB9C : 03...	Common Community	02-27-2015
Address 142.91.76.134	Common Community	02-27-2015

OSINT Framework

- <https://osintframework.com>
- Index of free recon tools!



Group Exercise

- Break into four equally sized groups
- Group 1:
 - State Farm
- Group 2:
 - Allstate
- Group 3:
 - Geico
- Group 4:
 - Progressive

Group Exercise

- Each group can use any of the tools we have previously discussed and answer the following questions
- You are free to visit any page that is indexed by Google.
 - Don't visit any page/subdomain that you found on DNS Dumpster that is not indexed by Google

Group Exercise Questions

1. What is their network IP range?
2. How many servers do they have online and from what countries?
3. How many subdomains were reported?
4. What interesting subdomains did you see reported?
5. Does it seem like they have a focus on security and why?
6. What are three other interesting things you uncovered?

Group Exercise Questions

- Each group should download reconexercise.docx from the Week 9 Canvas module for our course
- You have 15 minutes to answer the questions

Part II

Starting Litigation

Starting Litigation

- Formal litigation can start if enough information has been collected to pursue the case further
- Starting litigation involves:
 - A wide variety of documents to put the case on record with the court
 - Notification of potential parties and their duty to preserve evidence
 - Undertaking additional discovery in different forms

Legal Hold

- Legal hold: an order to preserve data in anticipation of litigation, an audit, a government investigation, or another matter
 - Similar terms: litigation hold, preservation letter, suspension order, and others
 - Destruction of evidence leads to sanctions

Legal Hold

- A legal hold should include the following items:
 - Locations of possible data
 - Possible types of relevant data
 - List of key custodians
 - List of things not to do (such as defragmenting, upgrading, and opening, moving, deleting, or overwriting files)
 - Contact person for questions

Preservation Letter Note

- These three questions in our group exercise last class were geared toward setting up a preservation letter/legal hold:
 - “Aside from Amazon, who are other third parties that may have relevant evidence?”
 - “What types of ESI would you include in a preservation letter to ensure relevant evidence isn’t deleted by Amazon?”
 - “What types of ESI would you include in preservation letters to ensure relevant evidence isn’t deleted by each of the third parties?”

Timing

- There are different rules for federal and state court regarding timing
- Federal court uses FRCP Rule 26 and declares that no party can seek discovery until the parties have met and conferred
- State courts may also require some sort of pre-discovery conference

Meet and Confer

- Parties must meet as soon as practical or before certain court deadlines
- The scope of the meeting is intended to encourage the parties to agree on litigation issues early in the case including:
 - The possibility for prompt settlement
 - If the case requires e-discovery
 - If so, discussions around developing a discovery plan
 - Other case resolution

Meet and Confer

- If both parties are using the same review platform:
 - **Cost sharing** can be used by hiring one vendor to process native files to TIF for all parties
- The meet and confer is the time to agree on access to computers or servers in question and on procedures to follow
- Search procedures should be established
 - Including keywords, custodians, date ranges, locations, and other matters to identify discovery as specifically as possible

Meet and Confer Note

- In the group exercise last week you spent most of the time meeting within your separate groups and preparing for the Meet and Confer
- The parts we covered that would actually happen in a real meet and confer would be the preservation proposal and discussion back and forth

Timing (Cont.)

1. Meet and Confer occurs
 - Discusses discovery plan and ESI Protocol
 - Must occur less than 21 days before scheduling conference with judge
2. Discovery Plan with ESI Protocol must be filed with court within 14 days of Meet and Confer
3. Scheduling conference occurs with judge

ESI Protocol

- One side will generally send the first draft and then each side modifies it back and forth
- Example ESI Protocol:
 - https://www.lieffcabraser.com/pdf/CLEF_3_Protocol.pdf

Next Steps

- After Discovery plan/ESI protocol is agreed to and the scheduling conference concludes, discovery can begin which can include:
 - Request for Admission
 - Interrogatories
 - Request for Production
 - Depositions

Interrogatory Note

- The below question from the last class exercise is an initial step in thinking about questions for an interrogatory
 - “What questions do you want to answer regarding the facts of the case relating to Amazon and potentially any third parties?”

Request for Production Note

- In both civil and criminal cases:
 - Subpoenas can be used in addition to RFPs or discovery motions
- A **subpoena duces tecum** is a command for a person to appear with certain evidence or to permit inspection of property

Request for Production Note

- When seeking discovery stored or transmitted over the Internet, the Stored Communications Act (SCA) can apply
- SCA lists five ways these records can be obtained:
 - Subpoena
 - Subpoena with previous notice to the subscriber or customer
 - Court order
 - Court order with previous notice to the subscriber or customer
 - Search warrant

Request for Production Note

- Main difference between a subpoena and a search warrant:
 - A subpoena commands the person to produce something
 - A search warrant permits law enforcement to search for and seize evidence if there's probable cause that a crime has been committed
 - Both can be challenged in court
- Motion to quash: A legal procedure used to reject, invalidate, or suppress evidence

Deposition Note

- Used to pin down facts, impeach a witness at trial, determine limits of witnesses' knowledge, etc.

Part III

Plaintiff's E-discovery Management

Plaintiff's E-discovery Management

- Before receiving discovery under an RFP:
 - A preliminary project management plan should be created
- Plan should identify:
 - The project team
 - Technology and tools
 - Quality control points and methods
 - Potential vendor support
 - Project budget and timeline
 - Case goals and objectives

Plaintiff's E-discovery Management

- Plan is updated as the case progresses
 - Having a well-documented discovery review process is helpful in defending against possible negative allegations
- When discovery is received:
 - A discovery or chain-of-custody log should be created
 - Includes details of items received such as date, type of media, media label data, etc.

Sampling and Metrics

- Sampling is done in order to:
 - Establish metrics
 - Identify unknown file types that might need special software or viewers
 - Verify the accuracy of search methods
 - Identify keywords to expand search parameters
- Metrics are gathered to:
 - Estimate the time and costs to manage, process, and review discovery
 - Verify the accuracy of searching and review procedures

Project Management Resources

- After metrics and sampling you should select:
 - Discovery tools
 - Vendors
 - Staff members

Part IV

E-discovery Organization

E-discovery Organization

- During the EDRM stage between collection and presentation:
 - Discovery is organized in notebooks or folder to keep track of chronology, issues, witnesses, experts, and trial outlines
- Review platforms offer features such as tags, folders, and sorting categories
 - Helps aid in organization
- Software offering more detailed or specific case organization:
 - LexisNexis CaseMap and TimeMap

Case Issues and Strategies

- Preliminary case issues and strategies are usually formed during the initial investigation but can be revised:
 - During the meet and confer
 - When discovery is received and reviewed
 - After depositions and other fact-finding probes
- When case issues move past 30 to 40:
 - Often means review staff can't organize large volumes of discovery correctly or consistently
 - Too many issues can lead to confusion and important issues get lost

Case Issues and Strategies

- Case issues should be well focused and grouped from the start
 - So that review staff can work with consistency
- Team leader should conduct quality control frequently to detect where problems might happen

Database Configuration

- The process of identifying and naming database fields takes the following factors into consideration:
 - Nature of the case
 - File types
 - Number and location of review team members, co-counsel, and other parties who need access
 - Security
 - Potential legal issues
- Databases have **key fields**
 - Unique numbers or identifiers used for image or data files

Database Configuration

- Security and access rights
 - Reviewers, remote staff or consultants, or other database users
- Team leader access to auditing and reporting tools
- Grant the lead attorney and review staff more limited security and access rights
 - They are mainly searching and tagging documents
 - Lead attorney may need rights to run reports

Quality Assurance

- For in-house e-discovery processing:
 - Quality assurance checks should be implemented at every phase
- Discovery inventories and chain-of-custody tracking are critical as well
- OCR quality, scanning resolution, and other factors should be check often
- Monitoring logs of database imports and exports
 - Ensures no discovery has been omitted inadvertently

Metrics Review

- Metrics are typically gathered in several stages of case processing
 - Often at the same time quality checks are done
- Metrics are used to evaluate time and budget concerns
- Without metrics:
 - Knowing whether certain stages of e-discovery could be completed before a case is ready to go to trial on time would be difficult
 - Most cases would go over-budget

Fact-Finding

- **Fact-finding:** the process of reviewing discovery to gather evidence that's relevant to case issues
 - Usually begins during prelitigation investigation but continues throughout the case life cycle
 - Can involve computer forensics examination
 - Used to review and organize e-mail, documents, audio, video, and other case materials
 - Includes organizing evidence by issue, case strategy, and relevant case law
 - Should include material that can support both sides of an argument

Search Concepts and Methods

- Different search concepts and methods are used, depending on factors such as:
 - Nature of the case and skills of the legal staff
- Early case organization and fact-finding are necessary to file pretrial motions and prepare effective trial presentations

Part V

Pretrial Motions & Trial

Pretrial Motions

- Pretrial motions are used to eliminate legal contentions or limit evidence before trial
 - Often result from facts discovered from evidence found during discovery review
 - There are usually court rules with specific deadlines for filing motions and responses
- **Motion to compel:** used to force the opposition to produce evidence previously requested but not delivered

Trial

- The book covers a few notes about trial presentation in the reading for this week
- In a few weeks we will go over how to testify and present evidence in court

Summary

- Early case evaluation includes prelitigation investigation, interviews, and preliminary discovery
- Formal litigation begins with filing a complaint in civil cases or an indictment in criminal cases
- A meet and confer is the time to agree on all technical and nontechnical discovery requirements
- Project management involves sampling and metrics
- The basis of e-discovery organization lies in case issues and strategies, which can often change and broaden as discovery is reviewed

Summary

- Fact-finding is the process of identifying evidence to support or refute case allegations
- Pretrial motions are used to eliminate legal contentions (dispositive) or limit evidence before a trial
- Judges have complete control over their courts and often have restrictions on placement or use of certain equipment in court and may limit or deny Internet access

Text References

- Phillips, A., Godfrey, R., Steuart, C., & Brown, C. (2014). E-discovery: An Introduction to Digital Evidence. Boston, MA: Cengage Learning.

Homework

- Review and understand this slide deck on Canvas
- Research paper due in four weeks on April 6th
prior to class start at 6:25pm
- We'll talk about the project in two weeks
- No class next week due to Spring Break

Questions???

